Approximating Euler's number

1

a 54.5982 b 0.3679 c 1.2214 d 8.2436

e 1.4715 f 0.0163

2

a $2.718\,280$ b $2.718\,281\,828$

3

a $2.012\,171$ b $2.013\,753$ c $0.001\,582$

Exponential expressions

4

a $7e^x$ b $9e^x$ c $3e^x$ d e^x

e $12e^{2x}+2e^x$ f $-5e^{2x}-8e^x$ g e^{x+2} h e^{5x}

i e^{5x} j e^{7x} k $2e^x-3$ l $\dfrac{e^{3x}+7e^{2x}}{6}$

5

a $e^{2x}+3e^x$ b $e^{3x}-9e^{2x}$

c $e^{2x}+7e^x+10$ d $e^{2x}-3e^x-10$

e $2+2e^x$ f $3+3e^{2x}$

g $2e+e^{2x}+e^{-2x}$ h $e^{2x}-2e^{-x}+e^{-4x}$

6

a $4\left(2e^x+3\right)$ b $4e^x(2e^x+3)$

c $4e^{3x}\left(2e^x-3\right)$ d $e^{4x}(8-e^{3x})$

e $\left(e^x-3\right)\left(e^x-2\right)$ f $(e^x+3)(e^x+3)$

g $\left(e^x+4\right)\left(e^x+2\right)$ h $(e^{2x}-5)(e^{2x}-2)$

Exponential graphs

7

a For $x>0$, the graph of $y=e^x$ will lie above the graph of $y=(2)^x$ and below the graph of $y=(3)^x$.

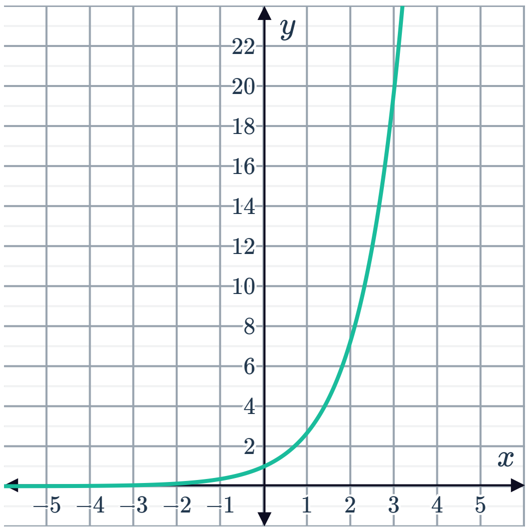
b For $x < 0$, the graph of $y = e^x$ will lie a[redacted]the graph of $y = (3)^x$ and below the graph of $y = (2)^x$.

8

a

x	-3	-2	-1	0	1	2	3
$f(x)$	0.05	0.14	0.37	1.00	2.72	7.39	20.09

b



9

a No

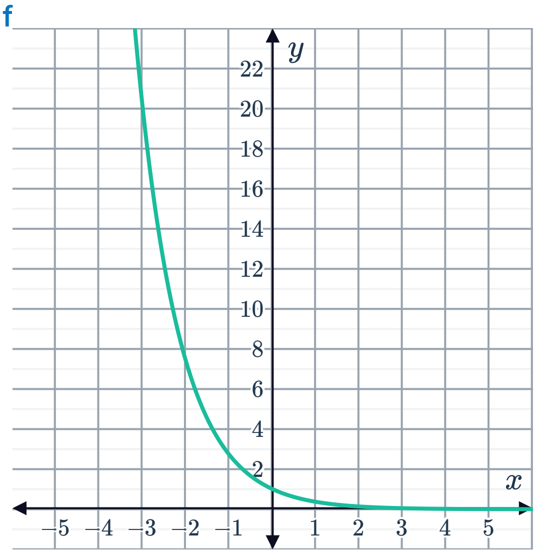
b **i** 0 **ii** ∞

c No

d 1

e 0

f



10 $(0, 1)$

11

a

x	$f(x)$
-5	-0.007
-4	-0.018
-3	-0.050
-2	-0.135
-1	-0.368
0	-1
1	-2.718
2	-7.389
3	-20.086
4	-54.598
5	-148.413
10	$-22\,026.5$

b No

c No

d Decreasing

e As x increases, the function decreases at a faster and faster rate.

Transformations of exponential graphs

12

a i No ii No iii No

b i No ii Yes iii Yes

c i Yes ii Yes iii Yes

d i No ii Yes iii Yes

13

a Increasing b Decreasing c Decreasing d Increasing

14

a A dilation by a factor of $\dfrac{1}{4}$ from the y-axis and a vertical translation of 1 unit upwards.

b A dilation by a factor of 5 from the x-axis and a horizontal translation of 4 units to the right.

c A dilation by a factor of 5 from the x-axis and a reflection about the y-axis.

15

a $y = e^{x+4} + 3$

b $y = e^{x+2} - 5$

c $y = 2e^{x} - 5$

d $y = -e^{x-2} + 3$

e $y = e^{\frac{x}{13}} - 2$

f $y = e^{3x} - 2$

16

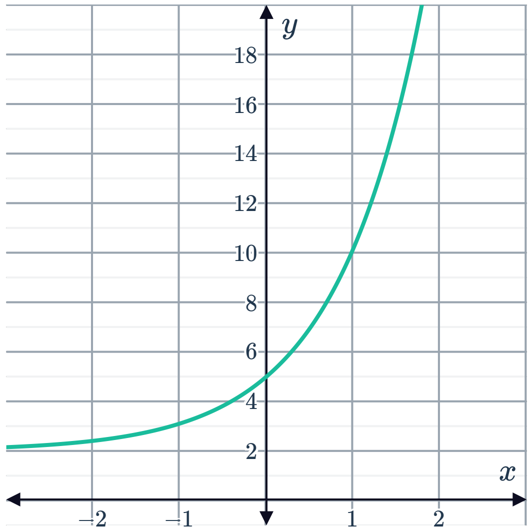
a

i $y = 3e^{x} + 2$

ii $y = 2$

iii 5

iv



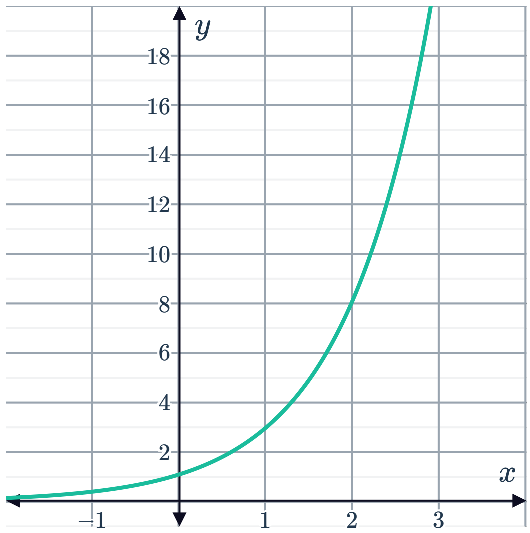
b

i $y = 3e^{x-1}$

ii $y = 0$

iii $\frac{3}{e}$

iv



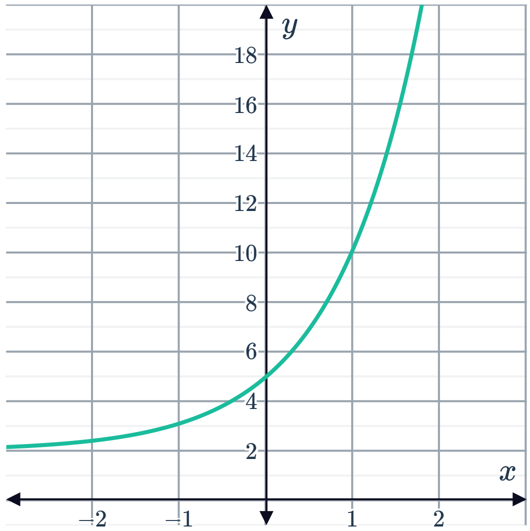
c

i $y = 3e^x + 2$

ii $y = 2$

iii 5

iv



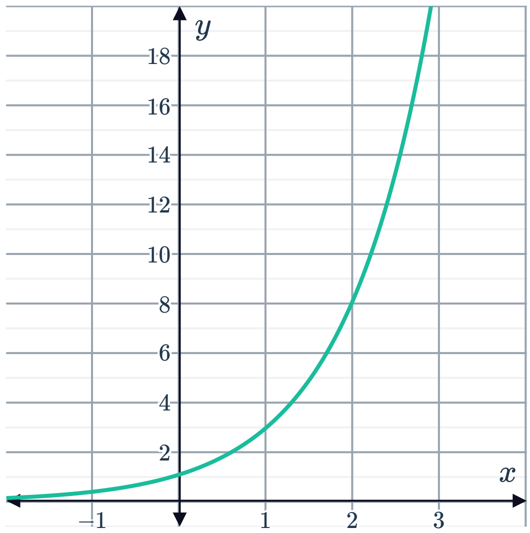
d

i $y = 3e^{x-1}$

ii $y = 0$

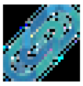
iii $\dfrac{3}{e}$

iv



e

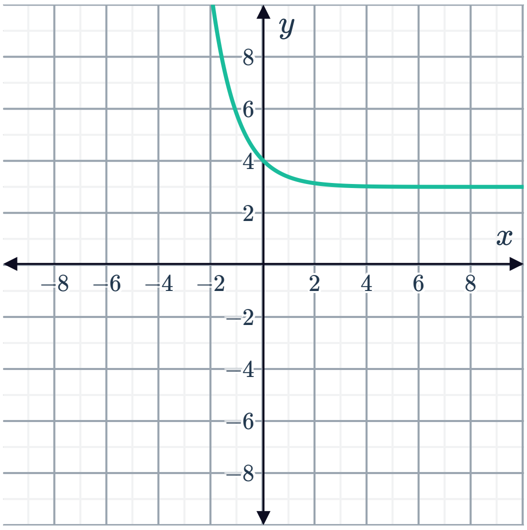
i $y = e^{-x} + 3$



ii $y = 3$

iii 4

iv



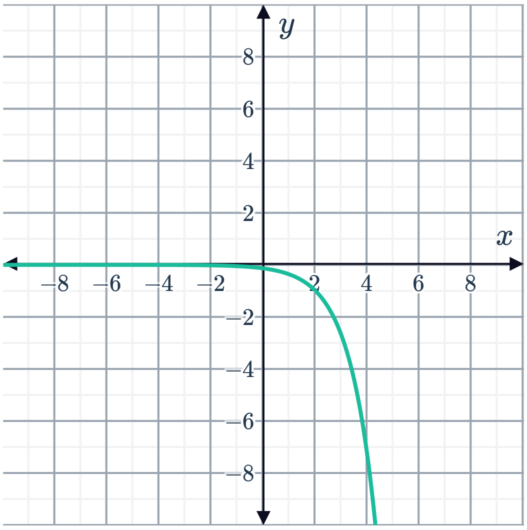
f

i $y = -e^{x-2}$

ii $y = 0$

iii $y = -\dfrac{1}{e^2}$

iv

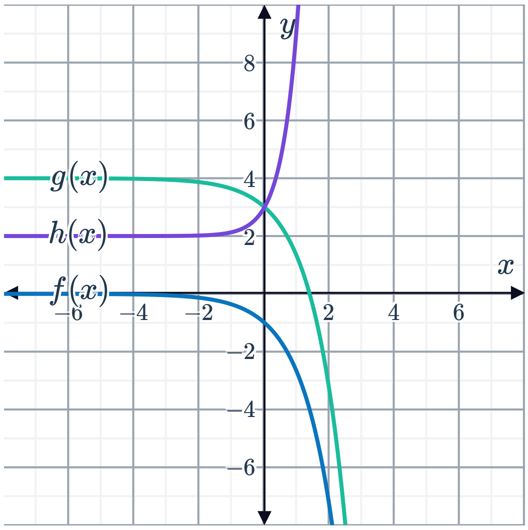


17

a

i $y = 0$ **ii** $y = 4$ **iii** $y = 2$

b



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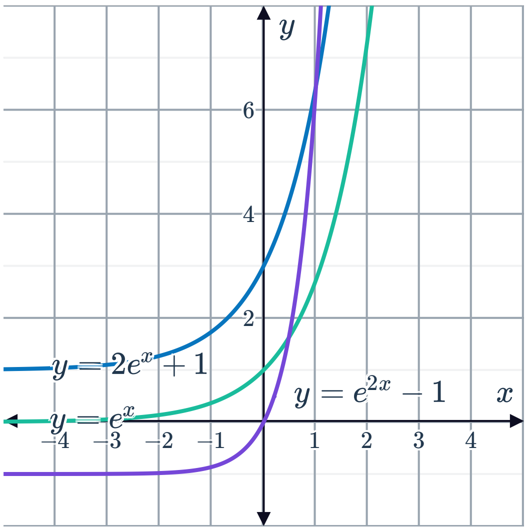

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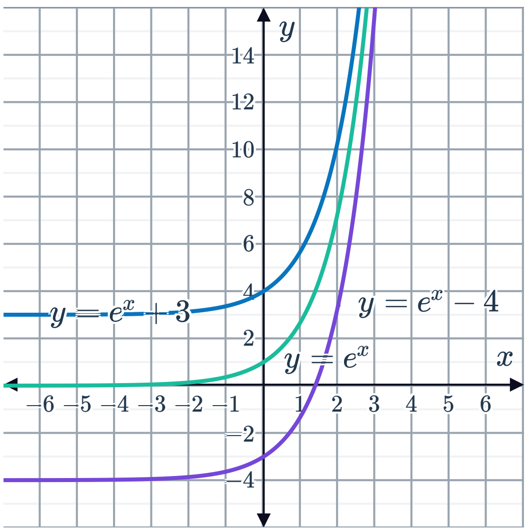

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